



NLP-BASED RESUME-JOB DESCRIPTION MATCHING

Name : Omkar Ishwar Shitole
 Class : AIEC-1
 Roll No. : 19
 Guide : Prof. Jyoti Gavhane

1. PROBLEM STATEMENT

- Recruiters manually screen many resumes, which is time consuming and inconsistent.
- Important skills and qualifications may be overlooked.
- There is a need for an automated system to compare a resume with a job description and rank candidates effectively.

2. OBJECTIVES

- ✓ Extract important skills and terms from resume and job description.
- ✓ Convert text into numerical form.
- ✓ Calculate similarity between resume and job description.
- ✓ Highlight matching skills and missing skills.
- ✓ Help recruiters save time and make better decisions.

CORE CAPABILITIES



TEXT UNDERSTANDING

Uses NLP to understand meaning in resumes and job descriptions.



TEXT COMPARISON

Calculates similarity score based on important skills and context.



SKILL MATCHING

Identifies matching skills and highlights missing skills.



FAST & EFFICIENT

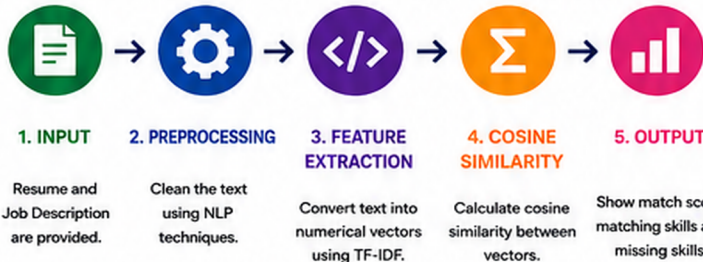
Automates resume screening and reduces manual effort.



DECISION SUPPORT

Helps recruiters choose the most suitable candidates.

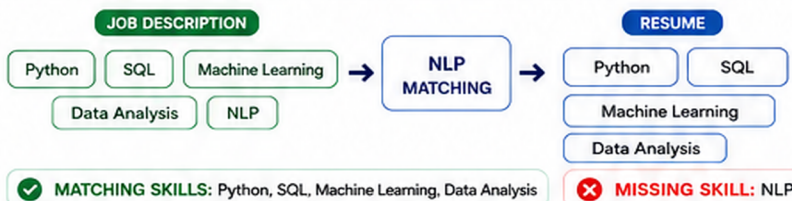
3. NLP WORKFLOW



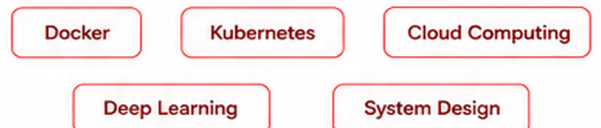
4. NLP TECHNIQUES USED

	Text Cleaning	Remove extra spaces, punctuation, numbers and special characters.
	Tokenization	Break text into smaller parts (tokens or words).
	Stop-word Removal	Remove common words like is, the, and, etc.
	Stemming / Lemmatization	Convert words to their root form (running → run).
	TF-IDF Vectorization	Convert text into vectors based on term importance.
	Cosine Similarity	Measure similarity between two vectors.

5. EXAMPLE (MATCHING & MISSING SKILLS)



MISSING SKILLS (EXAMPLE)



6. APPLICATIONS

- Recruitment and HR screening
- Applicant Tracking Systems (ATS)
- Career recommendation systems
- Skill gap analysis and upskilling

7. LIMITATIONS

- Different words can have similar meaning.
- Context and experience level are hard to capture.
- Similarity score is a support tool, not final decision.

8. CONCLUSION

NLP helps computers process and compare resumes with job descriptions. By applying NLP techniques, we can automatically match resumes with job descriptions, highlight key skills, and support better hiring decisions.

9. WHAT THE SYSTEM PROVIDES

